

Subject: Application for Working student robotics software engineering (m/f/d)

Dear Hiring Team,

I am excited to apply for the Working student robotics software engineering position at Pixel Robotics in Munich. I am currently pursuing my Master's degree in Electrical and Information Technology at Hochschule Darmstadt, specializing in Automation and Robotics. With my technical background in ROS2, SLAM, and autonomous navigation, along with my recent internship experience developing mobile robots, I am eager to contribute to Pixel Robotics.

As a Mobile Robots Development Intern at BÄR Automation GmbH, I gained hands-on experience optimizing navigation and localization systems for AMRs. Working with ROS 2 and Nav2, I supported autonomous movement and ensured VDA 5050 compliance. I compared and fine-tuned SLAM approaches, including SLAMToolbox and Cartographer with AMCL. By adjusting LiDAR and odometry parameters, I successfully reduced positional drift and improved accuracy. This aligns perfectly with your requirements for developing navigation algorithms and testing features in simulated and real-world deployments.

Academically, we developed indoor mapping and point cloud segmentation using a Hesai XT16 LiDAR on a UniTree quadruped robot.

Additionally, as a former Senior System Engineer and technical Scrum Master at Infosys, I bring a strong foundation in technical documentation and Agile/Scrum workflows. This corporate experience enables me to smoothly integrate into your team.

I am highly motivated to work alongside industry experts and learn while contributing to your innovative logistics solutions. I am a structured, proactive problem solver with a collaborative mindset, eager to tackle challenging engineering problems in your dynamic startup atmosphere.

Key Skills and Strengths:

- **Hands-on expertise** with ROS2, RVIZ, Gazebo, Nav2, Mapping, Localization, Linux OS, Git/GitHub
- **Proficiency in Python and C++** for algorithm development and optimization
- **Practical experience** testing and integrating software on real-world mobile robots

I am fluent in English (C1) and currently improving my German (A2.2). I am available to support your team with flexible hours and look forward to the opportunity to discuss how my passion for autonomous robotics can add value to Pixel Robotics.

Thank you for your time and consideration.

Best regards,
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